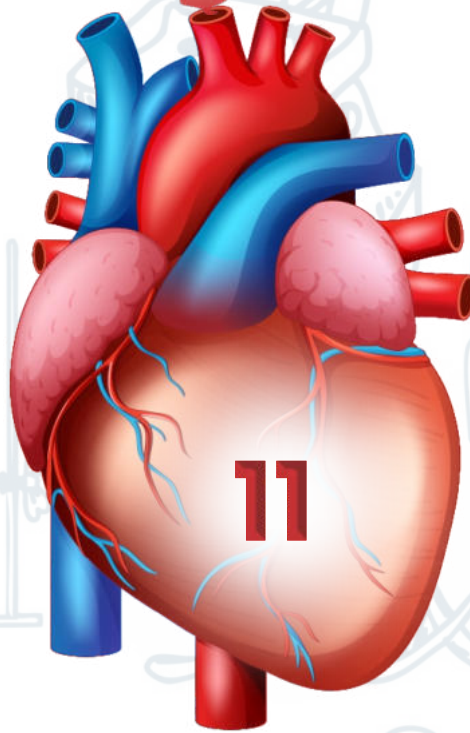
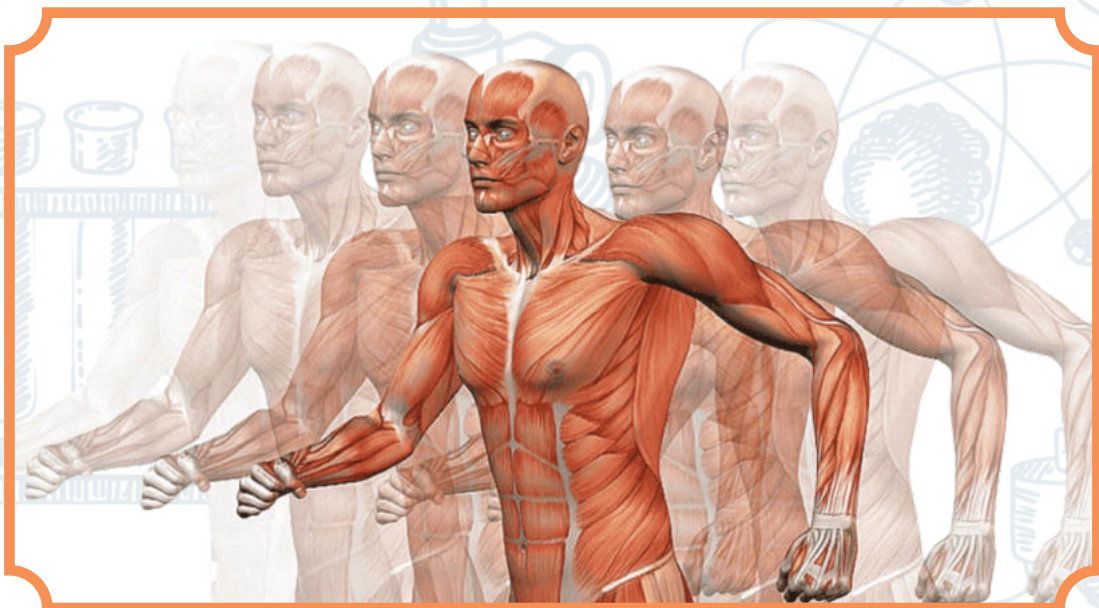


UNIT



HOLOZOIC NUTRITION



Holozoic Nutrition

Holozoic nutrition is a type of **heterotrophic nutrition** that involves the ingestion of solid or liquid food from the environment. It is commonly observed in **animals**, where food is consumed, digested, absorbed, assimilated, and finally egested. This process allows organisms to obtain energy and essential materials required for growth, repair, and other biological functions.

Key Stages of Holozoic Nutrition:

1. Ingestion:

- o The first step in holozoic nutrition is the **ingestion** of food, which involves taking in solid or liquid food from the environment.
- o Food may be consumed in various forms depending on the organism (e.g., through the mouth or specialized feeding structures).

2. Digestion:

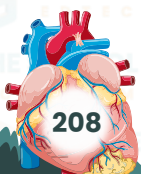
- o After ingestion, the food is broken down into simpler substances that can be absorbed by the body. This process is called **digestion**.
- o Digestion can occur either **extracellularly** (outside cells, typically in the digestive tract) or **intracellularly** (inside cells).
- o Enzymes play a crucial role in breaking down complex food molecules like carbohydrates, proteins, and fats into simpler forms like sugars, amino acids, and fatty acids.

3. Absorption:

- o Following digestion, the smaller food molecules are absorbed into the bloodstream or lymphatic system.
- o This absorption typically occurs in the **small intestine** in animals, where nutrients pass through the cell membranes into the bloodstream.

4. Assimilation:

- o Once absorbed, the nutrients are transported to the body cells, where they are used for growth, energy production, and repair. This process is called **assimilation**.
- o Nutrients are metabolized by cells to release energy or are used in the formation of new cell structures (proteins, lipids, etc.).



5. Egestion:

- o **Egestion** refers to the elimination of undigested food or waste materials from the body.
- o The waste products are typically expelled through the **anus** (in animals) or other excretory openings.

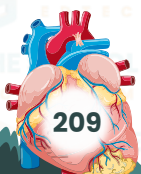
Significance of Holozoic Nutrition:

- **Energy Supply:** The primary function of holozoic nutrition is to provide the organism with **energy** for growth, maintenance, and reproduction.
- **Material for Growth:** It provides the essential **nutrients** (carbohydrates, proteins, lipids, vitamins, minerals) that are required for the development and repair of cells and tissues.
- **Metabolic Functions:** Through the process of assimilation, organisms can use absorbed nutrients in metabolic pathways that generate energy (like **cellular respiration**) or contribute to cellular structures and functions.

Ingestion, Digestion, Absorption, Assimilation, and Egestion

1. Ingestion:

- **Definition:** Ingestion is the process of **taking in food** into the cell (for unicellular organisms) or body (for multicellular organisms).
- **In Unicellular Organisms:** The food is taken in through the **cell membrane** via processes like phagocytosis (engulfing solid particles).
- **In Multicellular Organisms:** Food is ingested through a **specific opening** called the **mouth**.
- Examples:
 - o **Amoeba:** Engulfs food particles through its cell surface.
 - o **Humans and Animals:** Take in food through the mouth.



2. Digestion:

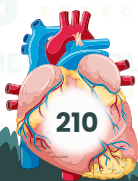
- **Definition:** Digestion is the process of **breaking down complex food** into **simple, soluble molecules** that can be absorbed and utilized by the body.
- **Types of Digestion:**
 - **Intracellular Digestion:** Occurs within the **cells** of unicellular organisms like **protozoa**. They ingest food directly into the cell and break it down using enzymes inside the cell.
 - **Extracellular Digestion:** Occurs outside the cells in the **digestive tract** or **gut** of multicellular organisms (e.g., humans, insects, and roundworms). **Digestive enzymes** are secreted into the gut to break down food into simpler substances that can be absorbed.
- **Mechanical and Chemical Digestion:**
 - **Chemical Digestion:** The breaking down of food using **enzymes**. Water is required to aid in the hydrolysis (breaking down with water) of food.
 - **Mechanical Digestion:** The physical breakdown of large food particles into smaller ones. This includes actions like **mastication** (chewing), **churning**, **grinding**, and **peristalsis** (muscular contractions in the digestive tract).

3. Absorption:

- **Definition:** Absorption is the process where **soluble molecules** from the digested food pass through the **digestive membranes** (e.g., intestinal lining) and enter the bloodstream or lymphatic system.
- **Purpose:** The absorbed nutrients are transported to various cells in the body for metabolic activities.
- **Example:** In humans, nutrients from the digested food are absorbed through the **walls of the small intestine**.

4. Assimilation:

- **Definition:** Assimilation is the process where the absorbed nutrients are utilized by **cells** during **metabolic processes** such as energy production, growth, and repair.
- **Examples:** The absorbed glucose may be used for **energy**, while absorbed amino acids are used to form new **proteins** in the cells.



5. Egestion:

- **Definition:** Egestion refers to the process of **eliminating** undigested food or **waste materials** from the body.
- **Example:** In humans, **feces** are excreted through the **anus**.

Intracellular and Extracellular Digestion

1. Intracellular Digestion (Unicellular Organisms):

- **Process:** Food is ingested and digested **inside the cell**.
- **Occurs in:** **Protozoa** and other unicellular organisms.
- **Mechanism:** The organism engulfs food particles via the **cell membrane**. The food is then broken down inside specialized compartments (like **food vacuoles**) by digestive enzymes.
- **Example:** **Amoeba** consumes food particles and digests them inside its cell.

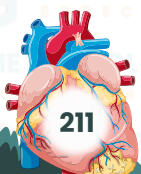
2. Extracellular Digestion (Multicellular Organisms):

- **Process:** Food is ingested into a **digestive tract** (gut), where it is digested **outside** the cells but within a specialized cavity.
- **Occurs in:** **Most animals** (e.g., humans, insects, and mammals).
- **Mechanism:** Food is ingested into the digestive system, where digestive **enzymes** are secreted by **glands** and **cells** into the **gut**. These enzymes break down food into simpler substances, which are then absorbed.
- **Example:** Humans ingest food through the mouth, where it is broken down in the stomach and intestines before nutrients are absorbed.

Sac-like and Tube-like Digestive Systems

1. Tube-like Digestive System:

- **Presence of Two Openings:**
 - o The digestive tract has two distinct openings: **mouth** (for food intake) and **anus** (for waste removal).
- **Found in:** Relatively **advanced groups** of organisms (e.g., humans, mammals, birds, etc.).



• Digestive Process:

- o **Food intake** and **waste removal** happen through separate openings.
- o **Mechanical and chemical digestion** both occur actively to break down food.
- o The digestive tract includes **associated glands** (e.g., salivary glands, pancreas, liver) that secrete digestive enzymes and fluids to aid in digestion.
- o **Digestive tract** has a **variable pH** (acidic or alkaline) depending on the complexity of food being processed.
- o **Enzymes:** There is a wide variety of enzymes that act on food components, facilitating effective digestion.
- o The digestive tract is well-supported by **muscles**, helping with peristalsis (the movement of food through the digestive system).

2. Sac-like Digestive System:

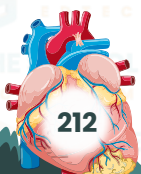
• Presence of Single Opening:

- o The digestive tract has a **single opening**, which functions as both the mouth and anus.

• Found in: Relatively **lower groups** of organisms (e.g., cnidarians, flatworms).

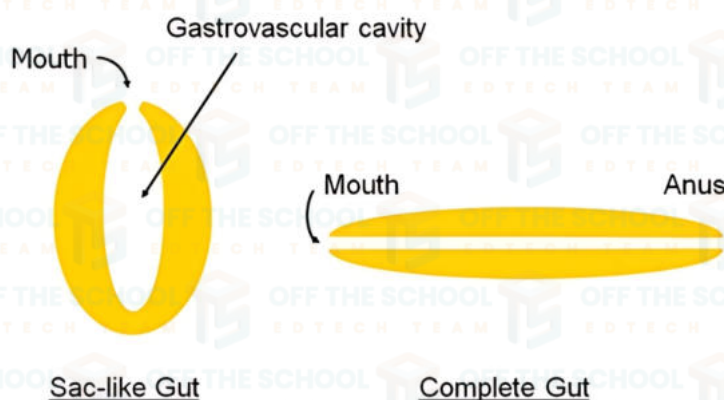
• Digestive Process:

- o **Food intake** and **waste removal** occur through the same opening.
- o The digestive system is **incomplete**, lacking specialized organs or glands (e.g., no salivary glands or pancreas).
- o **Mostly chemical digestion** occurs, as mechanical digestion is less developed.
- o The digestive tract does not have a variable pH, since the food is often simpler and easier to digest.
- o **Fewer enzymes** are present, and they mostly work on simpler food components.
- o The digestive tract is typically **not supported by muscles**, which limits movement and digestion efficiency.



Comparison:

Feature	Tube-like Digestive System	Sac-like Digestive System
Openings	Two (mouth and anus)	One (used for both food intake and waste removal)
Organisms	Advanced groups (e.g., humans, mammals)	Lower groups (e.g., cnidarians, flatworms)
Food Intake and Waste Removal	Separate openings for food intake and waste removal	Both occur through the same opening
Digestive Process	Mechanical and chemical digestion both occur actively	Mostly chemical digestion
Glands and Organs	Complete digestive system with associated glands	Incomplete system, no specialized organs
Enzymes	Numerous enzymes for various food components	Fewer enzymes for simpler food components
Muscle Support	Digestive tract well-supported by muscles	Digestive tract lacks muscle support
pH Variability	Digestive tract with variable pH	Digestive tract with little pH variation



Digestion in Amoeba:

Amoeba is a unicellular, eukaryotic organism that belongs to the **Protozoa** group. It is a **microphagous feeder**, meaning it feeds on minute food particles such as bacteria and other microscopic organisms.

Steps of Digestion in Amoeba:

1. Ingestion:

- o **Pseudopodia:** Amoeba captures food through its **pseudopodia** (temporary extensions of the cytoplasm). These extensions move in the direction of the food particles.
- o **Food Capture:** The pseudopodia surround and enclose the food particles, forming a **food cup** around them.
- o **Food Vacuole Formation:** Once the food is captured, the tips of the pseudopodia move towards each other, enclosing the food in a small spherical structure called the **food vacuole**. This marks the ingestion process.

2. Digestion:

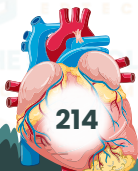
- o **Intracellular Digestion:** The food vacuole containing food particles moves within the cytoplasm. It is filled with various **hydrolytic enzymes** that break down the food.
- o **Enzymatic Action:** The enzymes break down the ingested food into simpler substances that can be absorbed by the cell. This digestion takes place inside the food vacuole.
- o **pH Level:** The pH within the food vacuole is maintained between 5.6 to 7.3, depending on the type of food.
- o **Intracellular Digestion:** The entire process of digestion occurs within the food vacuole, which is why it is termed intracellular digestion.

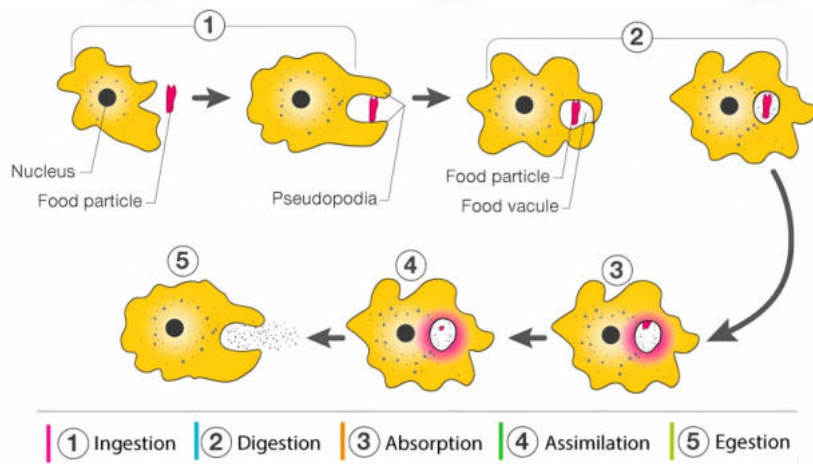
3. Absorption:

- o Once the food is digested, the **diffusible food particles** move out of the food vacuole into the cytoplasm.
- o The absorbed food particles are distributed to different parts of the cytoplasm where they can be used in metabolic processes.
- o The **digested food** may either be used to form new protoplasm or be broken down further to **release energy**.

4. Egestion:

- o After the absorption and utilization of nutrients, any undigested material is expelled from the cell through the process of **egestion**.





Digestion in planaria(Flatworm)

Planaria is a **free-living flatworm** belonging to the **Phylum Platyhelminthes**. It is commonly found in **wet places**, under rocks, near streams, canals, and in brackish water. Planaria has a **soft, elongated, unsegmented, and dorsoventrally flat body**.

Key Features of Digestive System in Planaria:

- The digestive system of Planaria is **incomplete** and is considered **sac-like**.
- The **mouth** is located at the end of a small, muscular tube known as the **pharynx**.
- The pharynx is attached to the intestine through a short esophagus.
- The intestine divides into three branches:

1. Anterior branch

2. Two posterior-lateral branches

- Each branch gives rise to numerous smaller branches, known as diverticula, which further divide into **fine, blind branches (ramifications)**.
- **No anus** is present; undigested food is expelled through the mouth.

Feeding and Digestion Process:

- **Carnivorous Diet:** Planaria feeds on **small larvae, insects**, and other small organisms.
- **Food Capture:** It captures prey by secreting **thick mucus** from the pharynx, which traps the prey.
- The prey is then **sucked up** through the mouth into the **pharynx**.

Extracellular Digestion:

- **Glandular cells** in the pharynx secrete **enzymes** that begin the process of **extracellular digestion**, breaking down the food as it moves through the pharynx and into the intestine.
- The **muscular action** of the body helps move food through the three branches of the intestine, breaking it down into smaller particles.

Intracellular Digestion:

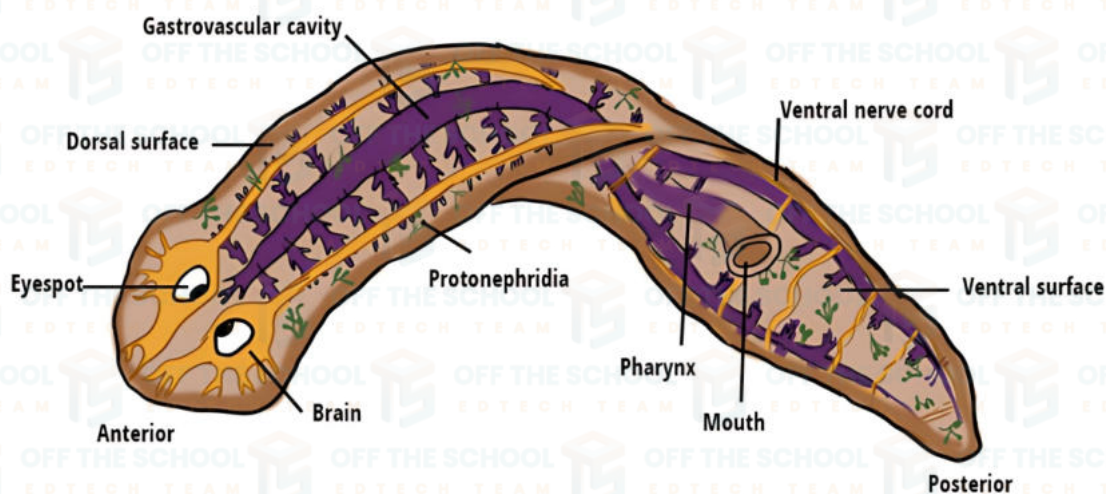
- As the food particles become smaller and simpler, they are absorbed by the cells lining the **gastrovascular cavity** (inside the intestine) via a process called endocytosis.
- Inside the **gastrovascular cavity cells**, the remaining digestion occurs, which is termed **intracellular digestion**.
- The digestion process in Planaria is **both extracellular and intracellular**.

Circulatory Role of the Intestinal System:

- The highly **branched intestine** helps distribute the digested food quickly to all cells of the body, functioning similarly to a **circulatory system**.
- The digested nutrients diffuse through the cells of the intestine and **mesenchyme cells**.

Egestion:

- **Undigested food** is expelled through the **mouth**, as Planaria lacks an anus.



Human Digestive System

The human digestive system is a **complete tube-like system**, also known as the **gastrointestinal tract (GI tract)**. It is designed to take in food, break it down into nutrients, absorb these nutrients, and expel waste products. The digestive system has two main openings: the **mouth** (for ingestion) and the **anus** (for egestion), forming a continuous tube.

Main Components of the Human Digestive System:

1. Mouth (Oral Cavity):

- o The digestive process begins here, where food is ingested.
- o **Salivary glands** secrete saliva, which contains enzymes (like amylase) to break down carbohydrates and lubricate the food for easier swallowing.

2. Pharynx:

- o The food passes from the mouth into the pharynx (throat), a passageway that connects the mouth to the esophagus.

3. Esophagus:

- o The food travels from the pharynx into the **esophagus**, a muscular tube that connects the throat to the stomach.
- o **Peristalsis** (rhythmic muscle contractions) moves food down the esophagus.

4. Stomach:

- o The food enters the stomach, where it is mixed with gastric juices (containing hydrochloric acid and enzymes like pepsin).
- o This helps break down proteins and churns food into a semi-liquid form called chyme.

5. Small Intestine:

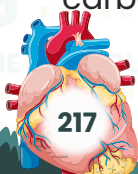
- o The chyme moves into the small intestine, which is the primary site for digestion and nutrient absorption.
- o The small intestine is divided into three parts:
 - **Duodenum:** Where bile from the liver and digestive enzymes from the pancreas aid in digesting food.
 - **Jejunum and Ileum:** These sections are responsible for nutrient absorption through villi, which are small, hair-like projections that increase surface area.

6. Liver:

- o The liver produces **bile**, which is stored in the **gallbladder**. Bile helps in the digestion and absorption of fats.
- o The liver also processes nutrients absorbed from the small intestine and detoxifies harmful substances.

7. Pancreas:

- o The pancreas secretes **digestive enzymes** into the duodenum that help break down carbohydrates, proteins, and fats.



8. Large Intestine:

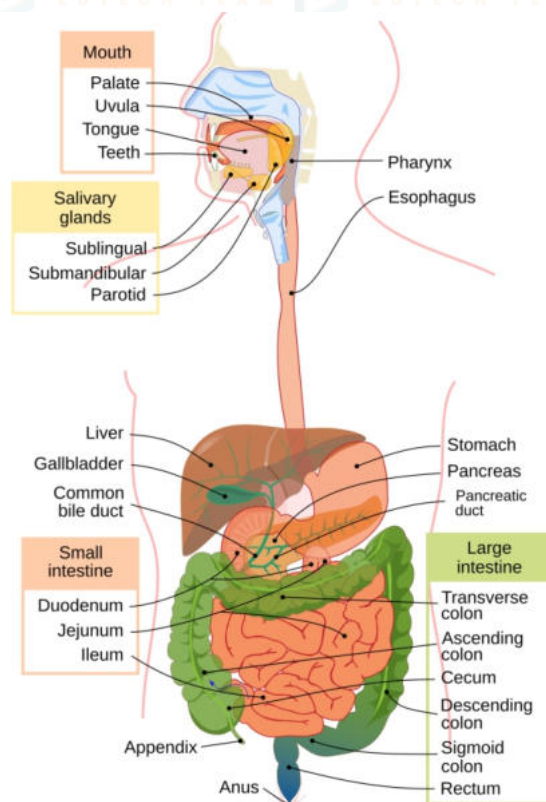
- o The remaining undigested food enters the large intestine, where **water** and **electrolytes** are absorbed.
- o The large intestine consists of the **cecum, colon, rectum, and anus**.
- o The **colon** absorbs water and salts, converting the liquid waste into solid stool.

9. Rectum and Anus:

- o The solid waste is stored in the **rectum** before being expelled through the **anus** during **defecation**.

Digestive Glands Associated with the System:

- 1. Salivary Glands:** Produce saliva that contains enzymes (amylase) for carbohydrate digestion.
- 2. Liver:** Produces bile to aid in fat digestion.
- 3. Pancreas:** Secretes digestive enzymes and bicarbonate to neutralize stomach acid.
- 4. Gallbladder:** Stores bile produced by the liver and releases it into the small intestine for fat digestion.



Oral Cavity:

The **oral cavity** is the first part of the digestive system and is responsible for the intake of food. It is formed by the upper and lower jaws and contains various structures that assist in the mechanical and chemical digestion of food.

1. Structure of the Oral Cavity:

- o **Teeth:** Used for cutting, tearing, and grinding food. The mouth contains both **deciduous (milk)** teeth and **permanent** teeth.
 - **Deciduous Teeth:** 20 teeth that are replaced by permanent teeth.
 - **Permanent Teeth:** 32 teeth in total, consisting of:
 - **8 Incisors** (for cutting),
 - **4 Canines** (for tearing),
 - **8 Premolars** (for grinding),
 - **12 Molars** (for crushing, no deciduous counterparts).
- o **Tongue:** A muscular organ that helps in mastication, swallowing, and speech. It is attached to the floor of the oral cavity via the **hyoid bone** and has **papillae** (projections with taste buds) on the surface.
- o **Salivary Glands:** There are three major pairs:
 - **Parotid glands** (near the ears),
 - **Submandibular glands** (under the lower jaw),
 - **Sublingual glands** (under the tongue).
 - These glands secrete **saliva**, which contains **amylase** to begin the breakdown of starch into maltose and **lysozyme** to help kill bacteria.

2. Teeth and Functions:

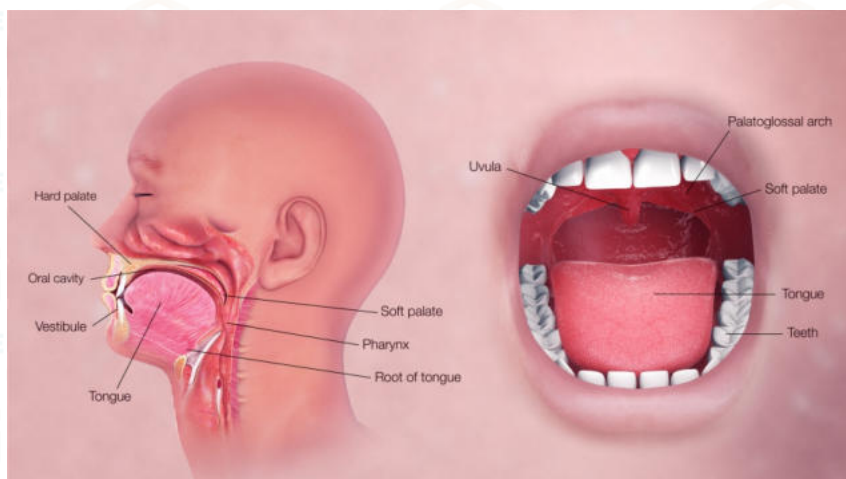
- o **Incisors:** For cutting and biting food.
- o **Canines:** For tearing and pulling food.
- o **Premolars and Molars:** For grinding and crushing food.

3. Tongue:

- o The tongue helps mix food with saliva, form a **bolus**, and aids in swallowing.
- o The **frenulum** is a fold of mucous membrane on the underside of the tongue.
- o The tongue also has sensory **papillae** for taste perception.

4. Salivary Glands:

- o Secrete saliva containing **mucin** (for lubrication) and **amylase** (for starch digestion).
- o Helps in the **chemical digestion** of food, particularly starch.
- o **1 to 1.5 liters of saliva** are produced daily.



Pharynx:

The **pharynx** is a muscular passage that connects the oral cavity to the esophagus and also to the nasal cavity.

1. Structure and Function:

- o It is connected anteriorly to the **oral cavity** and **nasal cavity**.
- o Posteriorly, it connects to the **esophagus** (for food passage) and **larynx** (for air passage).
- o It allows both food and air to pass through and plays a crucial role in **swallowing**.
- o The **Eustachian tube** connects the pharynx to the middle ear, helping to equalize air pressure on the eardrum.

2. Muscles of the Pharynx:

- o **Pharyngeal constrictor muscles** help in the process of swallowing.

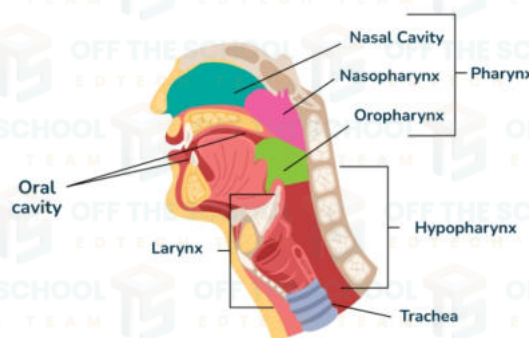
Swallowing (Deglutition):

After food is chewed and mixed with saliva in the oral cavity, it forms a soft, lubricated ball-like structure called **bolus**.

1. Swallowing Process:

- o The **tongue** pushes the bolus to the back of the mouth into the **pharynx**.
- o The process is **voluntary**, and once the bolus reaches the pharynx, it becomes **involuntary**.

Diagram of Pharynx



- o The **soft palate** elevates to prevent food from entering the nasal cavity.
- o The **larynx** moves upward and forward to allow the **epiglottis** to close over the trachea, diverting the bolus into the **esophagus**.

Peristalsis:

• **Definition:** Peristalsis is a wave-like, **involuntary** rhythmic movement that involves the **contraction and relaxation** of the **smooth muscles** in the walls of the **alimentary canal** (digestive tract).

• **Function:**

- o This movement **propels food forward** through the digestive system, from the **pharynx** to the **esophagus**, and all the way to the end of the digestive tract.
- o It also occurs in other tube-like structures in the body, such as in blood vessels and

• Mechanism:

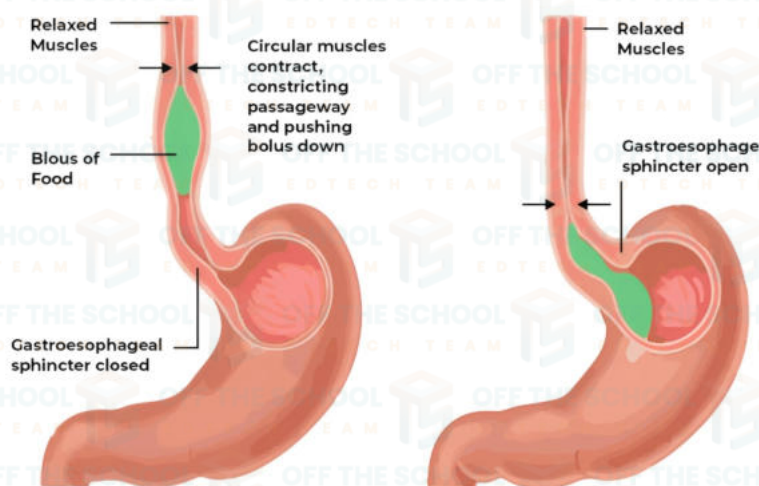
- o Smooth muscles contract behind the food (in the direction of movement).
- o The muscles relax in front of the food, allowing it to move forward.

• Occurrence:

- o **After eating a large meal**, peristalsis helps to move the food along the digestive tract, ensuring that digestion and absorption occur efficiently.

• Importance:

- o Peristalsis is essential for the **transport** of food, waste, and liquids through the digestive system, aiding in **digestion** and **absorption**.



Stomach Structure and Function:

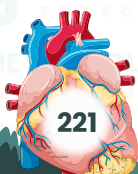
• Regions of the Stomach:

- 1. Fundus:** The uppermost portion.
- 2. Corpus (Body):** The central part of the stomach, primarily responsible for storing food and mixing it with gastric juices.
- 3. Pylorus:** The lower portion that empties the gastric contents into the small intestine.

• Stomach's Function:

- o The stomach acts as a **reservoir** for ingested food, adapting to the varying volumes of food consumed by **relaxing** and **expanding**.
- o It contains **gas bubbles**, particularly after meals, to aid in digestion.
- o It is responsible for **digesting** food, producing **gastric juices**, and pushing the contents to the small intestine for further digestion and absorption.

• Stomach Wall Structure: The stomach wall consists of **four layers**:



0. Mucosa:

- The innermost layer, which is **thicker** than in other parts of the gastrointestinal tract.
- Contains **tubular glands** that secrete gastric juices.

1. Submucosa:

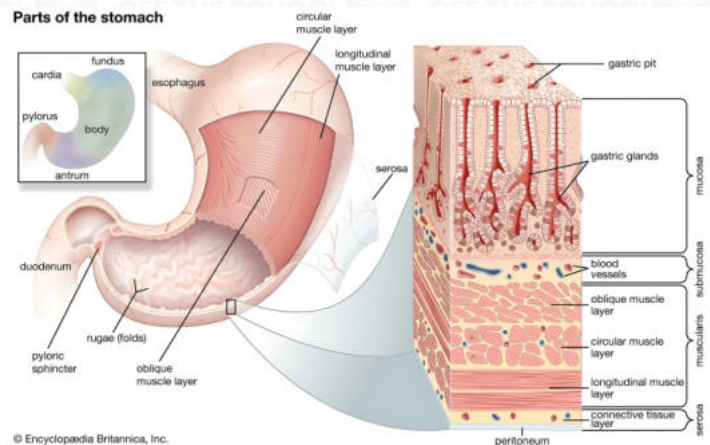
- Lies beneath the mucosa.
- Contains **blood vessels, nerves**, and other support structures.
- **Glands** are absent in this layer.

2. Muscularis Externa:

- This layer is composed of **three smooth muscle layers**:
- **Inner oblique layer**
- **Middle circular layer**
- **Outer longitudinal layer**
- These muscles aid in the **mechanical digestion** by churning food.

3. Serosa:

- The outermost layer, which is **thin** and secretes **lubricating fluid** to reduce friction from muscle movement.



Structure of the Stomach Mucosa and Gastric Glands:

• Empty, Contracted Stomach:

- o The **mucosa** of the stomach is thrown into **longitudinal folds** known as **rugae** due to the contraction of the muscular layer.
- o The rugae help the stomach **expand** when food enters and allows for better mixing and digestion.

• Mucosal Surface:

- o The surface is lined by **epithelium**, and contains **funnel-shaped invaginations** called gastric pits.
- o These pits are continuous with the tubular **gastric glands** and contain **mucus** and **bicarbonate ions**, which **protect the stomach lining** from its own acidic secretions (hydrochloric acid or HCl).

Gastric Glands and Their Secretions:

1. Types of Cells in the Gastric Glands:

o Mucous Cells:

- Found in the surface epithelium and gastric pits.
- They secrete **mucus** and **bicarbonate ions**, which help **protect** the stomach from the damaging effects of **Hcl**.

o Parietal (Oxyntic) Cells:

- Located in the **fundus** and **corpus** regions of the stomach.
- They secrete **hydrochloric acid (HCl)** and **intrinsic factor**.
- **HCl** aids in digestion and creates an acidic environment for enzymes to function.
- **Intrinsic factor** is crucial for vitamin B12 absorption in the small intestine and for the production of **red blood cells**.

o Chief (Zymogen) Cells:

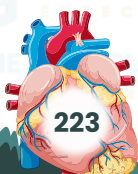
- Found in the gastric glands.
- They secrete **pepsinogen**, an inactive enzyme that is converted into **pepsin**, which helps in protein digestion.

2. Pyloric Glands:

- o Found in the **pylorus** (distal region of the stomach).
- o These glands secrete **mucus** and the hormone **gastrin**.
- o Gastrin is released in response to:
 - **Stomach distention** (when food enters).
 - **Increase in stomach pH** (e.g., from protein-rich food, which decreases acidity).
- o **Gastrin** stimulates the production of **gastric acid** and aids in digestion.

3. Endocrine Cells:

- o These cells are scattered throughout the stomach.
- o They secrete **serotonin**, which helps **regulate gastric acid secretion**.



Functions of Gastric Secretions:

- **Hydrochloric Acid (HCl):**

- o Provides an acidic environment in the stomach that aids in the activation of digestive enzymes like **pepsin**.

- **Mucus:**

- o Protects the stomach lining from the corrosive effects of acid and enzymes.

- **Pepsinogen:**

- o The inactive form of **pepsin**, which is activated in the acidic environment to break down proteins.

- **Intrinsic Factor:**

- o Necessary for the absorption of **vitamin B12** in the small intestine.

Physiological Processes of Mechanical and Chemical Digestion in the Stomach:

1. Initiation of Digestion:

- **Food Intake:**

- o As soon as food enters the stomach, both **mechanical** and **chemical** digestion begin simultaneously.

- o The **anticipation** of food (through sight, smell, taste, sound) stimulates the **vagus nerve**, which sends signals to the brain.

- o **Distention of the stomach**, particularly by proteins, also triggers the release of the hormone **gastrin** into the bloodstream.

2. Role of the Vagus Nerve and Gastrin:

- **Vagus Nerve:**

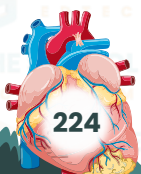
- o Part of the **parasympathetic nervous system**, responsible for regulating many bodily functions, including digestion.

- o It triggers **gastric glands** to release **gastric juice** into the stomach.

- **Gastrin:**

- o Released in response to stomach distention.

- o Stimulates further secretion of **gastric juice**.



3. Gastric Juice and its Functions:

- **Components of Gastric Juice:**

- o Contains **hydrochloric acid (HCl)** and **pepsinogen**.
- o **HCl** softens the food, kills harmful microorganisms, and creates an acidic environment for enzyme activity.
- o **Pepsinogen** is converted into its active form, **pepsin**, by the acidic environment.

- **Pepsin Function:**

- o **Pepsin** breaks down proteins into **peptides** (shorter chains of amino acids).

4. Protection of the Stomach:

- The stomach lining is protected from its own acidic secretions by a **thick layer of mucus** that contains **bicarbonate ions**.
- o These ions neutralize the acidity in the stomach lining, maintaining a pH around **1.5 to 2.5**.

5. Mechanical Digestion:

- **Muscular Layers of the Stomach:**

- o The stomach has three layers of **smooth muscles**:

- **Inner oblique layer**

- **Middle circular layer**

- **Outer longitudinal layer**

- o These muscles contract in **peristaltic movements**, which:

- **Churn** the food,

- **Mix** it with gastric juices,

- **Transport** the food to the **duodenum** (small intestine).

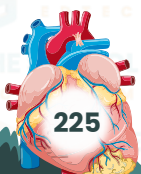
- **Food Churning:**

- o The food becomes a **paste-like substance** (chyme) and is gradually transported to the duodenum, increasing surface area for further digestion.

6. Absorptive Role of the Stomach:

- While the stomach's primary function is digestion, it also plays a **minor role in absorption**:

- o Absorbs small amounts of **water, glucose, simple sugars, amino acids, and fat-soluble substances** (e.g., alcohol, certain medications).



Small Intestine:

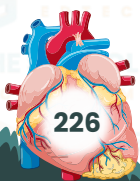
The small intestine is a long, coiled tube, approximately **6 meters** long and **3-4 cm** wide, responsible for completing digestion and absorbing nutrients.

Divisions of the Small Intestine:

- 1. Duodenum:** First part, about **30 cm** long.
- 2. Jejunum:** Middle part, approximately **2.5 meters** long.
- 3. Ileum:** Last part, about **3.6 meters** long.

Duodenum:

- **Location:** Connected to the stomach via the pyloric sphincter.
- **Functions:**
 - **Receives digestive juices** from the liver (bile) and pancreas (pancreatic juice).
 - **Bile:**
 - Produced by the liver, stored in the gall bladder.
 - Released into the duodenum in response to **cholecystokinin** (a peptide hormone).
 - Functions to **emulsify fats** (breaking them into smaller droplets) but does not **contain enzymes**.
 - **Pancreatic Juice:**
 - Contains **trypsinogen**, which is activated into **trypsin** by **enterokinase** (secreted by duodenum).
 - **Trypsin** and **chymotrypsin** break down proteins into short-chain peptides.
 - **Amylase** breaks down starches into **maltose**.
 - **Lipase** breaks down emulsified fats into **fatty acids** and **glycerol**.
 - **Brunner's Glands:** Secrete an alkaline mucus to neutralize stomach acid, protecting the duodenum from damage.
 - **Cholecystokinin** also stimulates the pancreas to release **pancreatic juice**.
 - **Secretin:** A hormone that promotes secretion of **water and bicarbonate** ions from the pancreas.



Jejunum:

- **Location:** Middle section of the small intestine.
- **Length:** About **2.5 meters**.
- **Function:**
 - **Completion of digestion.**
 - Secretes enzymes like **maltase, sucrase, lactase, and peptidase**, which break down **maltose, sucrose, lactose, and peptides** into simple sugars and amino acids.
 - **Nucleotidase** breaks down nucleotides into **nucleosides**.
 - **Cellulose** remains undigested here.

Ileum:

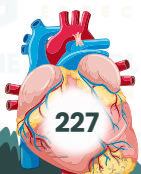
- **Location:** Last part of the small intestine.
- **Length:** About **3.6 meters**.
- **Function:**
 - Absorbs **Vitamin B12, bile salts**, and any remaining digestive products not absorbed by the jejunum.

Structure of the Small Intestine:

- **Plicae Circulares:** Large folds in the walls of the small intestine that increase surface area.
- **Villi:** Finger-like projections arising from the folds.
- **Microvilli:** Tiny hair-like projections on the epithelial cells of the villi, further increasing surface area.
 - Together, these structures increase the **surface area for absorption**.

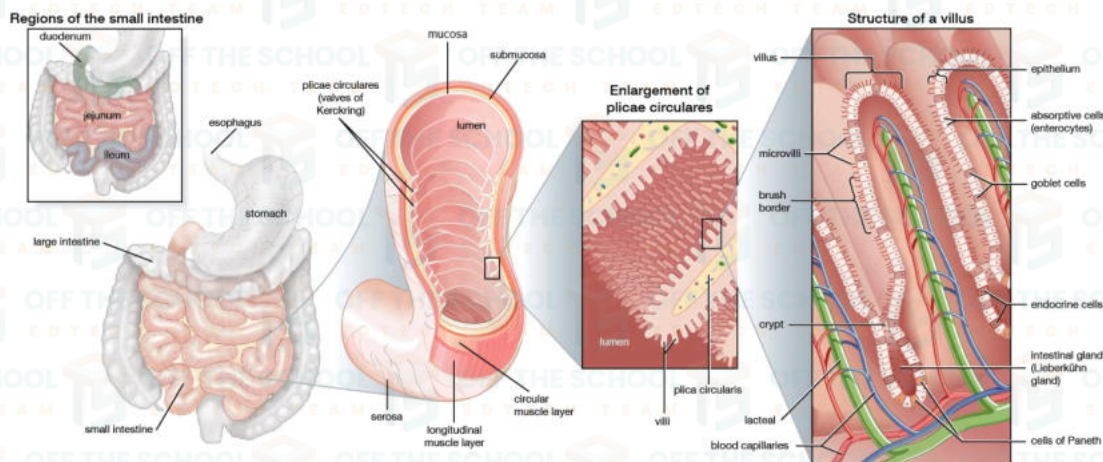
Absorption in the Small Intestine:

- **Villi** contain a network of **blood capillaries** and **lacteals** (lymph vessels).
 - Monosaccharides and **amino acids** are absorbed by blood capillaries through **diffusion** or **active transport**.
 - **Fatty acids** and **glycerol** are absorbed by **lacteals**.
 - **Lacteals** merge into larger lymph vessels and transport nutrients to the **thoracic duct**, which drains into the **subclavian vein**.
 - Blood capillaries carrying absorbed nutrients merge into the **hepatic portal vein**, which transports nutrients to the liver for storage, metabolism, and distribution.



Contractility of Villi:

- Villi can contract and relax due to the smooth muscle, bringing them into close contact with the food for efficient absorption.



Large Intestine:

The **large intestine** is the last segment of the human digestive tract, responsible for water reabsorption, vitamin synthesis, and the formation of feces.

General Features:

- **Length:** Approximately **1.5 meters** long, shorter compared to the small intestine.
- **Main Functions:**
 - **Reabsorption of water** and some nutrients.
 - **Synthesis of vitamins**, including some B-vitamins and vitamin K by bacteria.
 - **Reabsorption of electrolytes** (e.g., sodium, chloride).
 - **Formation of feces** from undigested food.
 - Mucous secretion: Helps in lubrication, aiding in the smooth passage of contents through the intestine.
 - Lacks **villi**, which are present in the small intestine.

Regions of the Large Intestine:

1. Caecum:

- Located at the junction of the small and large intestines.
- Positioned in the lower-right abdomen.
- The **ileum** (last part of the small intestine) connects to the **caecum**.
- **Appendix:** A blind, tube-like structure (~18 cm long) attached to the caecum, known as the **vermiform appendix**. Its function is uncertain but is thought to have an immune function.

2. Colon:

o Divided into several parts:

- **Ascending colon:** Travels up the right side of the abdomen.
- **Transverse colon:** Runs across the abdomen.
- **Descending colon:** Moves downward along the left side.
- **Sigmoid colon:** S-shaped section that leads to the rectum.

o **Main Functions:**

- **Absorption of water** and electrolytes.
- **Absorption of vitamins** (e.g., vitamin B1, B12, niacin, and vitamin K) produced by symbiotic bacteria, which are essential for the body.
- **Fermentation of undigested carbohydrates** by gut bacteria to produce gases and short-chain fatty acids, which are absorbed for energy.

3. Rectum:

o Last section of the large intestine, approximately **13 cm** long.

o **Function:** Temporarily stores feces.

o **Feces formation:** Fecal material moves from the sigmoid colon into the rectum.

o When the rectum is full, it sends an impulse to the brain signaling the need to defecate.

4. Anal Canal:

o **Internal sphincter:** Composed of smooth muscle and is **involuntary**, controlling the release of feces.

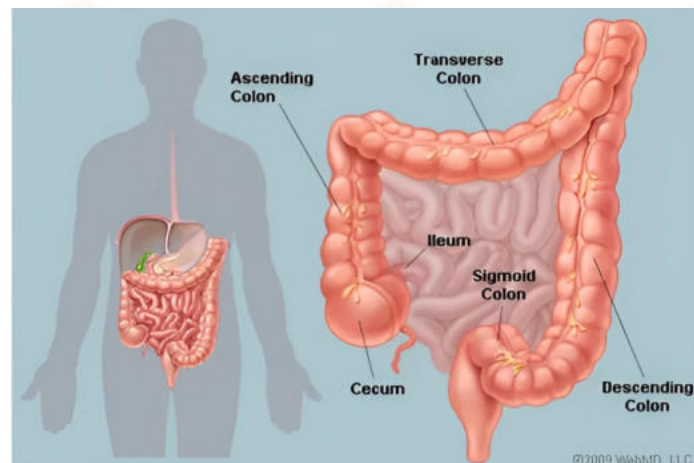
o **External sphincter:** Composed of skeletal muscle and is **voluntary**, allowing control over the timing of defecation.

o Defecation involves the **relaxation** of the internal sphincter and the **contraction** of the external sphincter.

Defecation Process:

• In **adults**, defecation is a controlled process involving the contraction of the **rectum** and the relaxation of the internal sphincter, followed by the voluntary contraction of the external sphincter.

• In **infants**, defecation is involuntary due to the lack of voluntary control over the external sphincter.



ROLE OF ACCESSORY GLANDS IN DIGESTION

Accessory glands are those that secrete essential digestive substances into the alimentary canal through specific ducts. These secretions help in the breakdown, digestion, and absorption of food.

1. Salivary Glands

- **Type:** Exocrine glands.
- **Number:** Three pairs – Parotid, Submandibular, and Sublingual glands.
- **Secretion: Saliva**, containing:
 - **Salivary amylase (ptyalin)** – breaks down starch into maltose.
 - **Mucus** – lubricates food for easier swallowing.
- **Stimulation:** Triggered by the sight, smell, or presence of food.

2. Liver

- **Location:** Upper right side of the abdominal cavity, just below the diaphragm.
- **Structure:** Four lobes – right, left, caudate, and quadrate; made up of tiny units called **lobules** containing hepatocytes (liver cells).

• Main Functions:

a. Secretion

- **Bile production:**
 - Made by hepatocytes, stored in the gall bladder.
 - Contains bile salts (sodium glycocholate, taurocholate), bile pigments (bilirubin, biliverdin), and cholesterol.
 - Helps in **emulsification of fats**.

b. Metabolism & Homeostasis

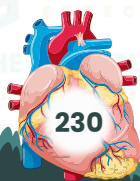
- **Glucose** → **Glycogen** (by insulin) and **Glycogen** → **Glucose** (by glucagon).
- **Deamination of amino acids** – forming urea and uric acid.
- **Formation of non-essential amino acids.**

c. Synthesis

- **Plasma proteins, blood clotting factors**, and **vitamin A from carotene**.

d. Breakdown & Detoxification

- **Breaks down worn-out RBCs.**
- **Kupffer cells:** phagocytose pathogens and debris.
- **Detoxifies drugs** (e.g., alcohol) and inactivates hormones (insulin, cortisol, etc.).



e. Storage

- **Vitamins:** A, D, E, K, B12.
- **Minerals:** Iron, copper.
- **Other B-complex vitamins:** Riboflavin, niacin, pyridoxine, folic acid.

f. Heat Production

- High metabolic activity makes it a **major heat-producing organ**.

Liver Disorders

- **Hepatitis:** Inflammation of the liver caused by viruses, drugs, toxins, or autoimmune disorders.
 - Symptoms: Fatigue, jaundice, dark urine, pale stool, abdominal pain.
- **Jaundice:** Yellowing of skin and eyes due to excessive **bilirubin** from broken-down RBCs.

3. Pancreas

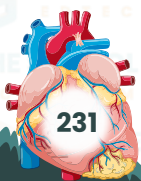
- **Location:** Behind the stomach, along the curve of the duodenum.
- **Length:** 12–15 cm.
- **Dual Function:**

a. Exocrine Function

- Secretes **pancreatic juice** via the **pancreatic duct**, which joins the **common bile duct** at the **hepatopancreatic ampulla** (controlled by **sphincter of Oddi**).
- Contains enzymes:
 - **Amylase** – digests starch to maltose.
 - **Lipase** – breaks fats into fatty acids and glycerol.
 - **Trypsinogen, chymotrypsinogen** – digests proteins after activation.

b. Endocrine Function

- (involves insulin and glucagon from the **Islets of Langerhans**.)



1. Ulcer (Peptic Ulcer):

- **Definition:** A painful sore in the lining of the stomach or small intestine caused by damage from digestive acids.
- **Causes:**
 - o Helicobacter pylori infection.
 - o Overuse of medications like aspirin or NSAIDs.
 - o Excess alcohol consumption or spicy food.
- **Symptoms:** Sharp abdominal pain, nausea, vomiting, and loss of appetite.
- **Treatment:** Antibiotics and proton-pump inhibitors.

2. Food Poisoning:

- **Definition:** Illness caused by consuming contaminated or spoiled food.
- **Causes:** Bacteria (Staphylococcus), viruses, parasites.
- **Symptoms:** Stomach cramps, diarrhea, vomiting, fever.
- **Treatment:** Hydration with electrolyte solutions; antibiotics if bacterial.

3. Dyspepsia (Indigestion):

- **Definition:** Discomfort or pain in the upper abdomen after eating.
- **Symptoms:** Bloating, nausea, heartburn, abdominal fullness.
- **Cause:** Acid reflux, overeating, stress, or fatty foods.

4. Obesity:

- **Definition:** Excessive accumulation of body fat ($\text{BMI} > 30 \text{ kg/m}^2$).
- **Causes:** Genetic factors, poor diet, lack of physical activity.
- **Risks:** Diabetes, heart disease, hypertension, certain cancers.
- **Treatment:** Long-term lifestyle change—healthy diet, regular exercise.

5. Anorexia Nervosa:

- **Definition:** A psychological disorder marked by an intense fear of weight gain and self starvation.
- **Symptoms:** Severe weight loss, loss of appetite, refusal to eat, induced vomiting.
- **Mostly affects:** Adolescents and young adults, especially females.

6. Bulimia Nervosa:

- **Definition:** A psychological disorder involving binge eating followed by purging (vomiting or laxatives).
- **Symptoms:** Fluctuating weight, dehydration, electrolyte imbalance.
- **Complications:** Kidney damage, heart issues, gastrointestinal problems.

